



HEILONGJIANG BIANFUSHAN WIND POWER PROJECT

Document Prepared by Longyuan (Beijing) Carbon Asset Management
Technology Co., Ltd.

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CONTENTS

1 PROJECT DETAILS	1
1.1 Summary Description of the Implementation Status of the Project.....	1
1.2 Sectoral Scope and Project Type	1
1.3 Project Proponent	1
1.4 Other Entities Involved in the Project	2
1.5 Project Start Date	2
1.6 Project Crediting Period	2
1.7 Project Location	2
1.8 Title and Reference of Methodology	3
1.9 Participation under other GHG Programs	4
1.10 Other Forms of Credit	4
1.11 Sustainable Development Contributions	5
2 SAFEGUARDS	8
2.1 No Net Harm	8
2.2 Local Stakeholder Consultation	8
2.3 AFOLU-Specific Safeguards	9
3 IMPLEMENTATION STATUS	9
3.1 Implementation Status of the Project Activity	9
3.2 Deviations	10

3.3 Grouped Projects.....	10
4 DATA AND PARAMETERS	11
4.1 Data and Parameters Available at Validation	11
4.2 Data and Parameters Monitored	11
4.3 Monitoring Plan	16
5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS	19
5.1 Baseline Emissions	19
5.2 Project Emissions	26
5.3 Leakage	26
5.4 Net GHG Emission Reductions and Removals	26
APPENDIX1: <SUPPORTING EVIDENCE>	28

1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

The Project is a newly-build wind plant with total installed capacity of 49.5MW (33*1500kW). The project generated electricity from wind resources using advanced wind power generation technology on a commercial basis and delivered the electricity to the Northeast China Power Grid (NECPG). It is estimated that the feed-in electricity to the NECPG is approximately 121,625MWh per year with 2457h designed annual operation hours. The Project activity achieved greenhouse gas (GHG) emission reductions by avoiding CO₂ emissions from the business-as-usual scenario electricity generated by those fossil fuel-fired power plants connected into NECPG. It is estimated that annual emission reductions are 125,030tCO₂e per year.

The project was implemented in consistent with the registered PD. The project began to construct in 04-Jun-2009 and started commissioning on 25 December 2009.

This is the 2nd VCS monitoring period from 01-Jan-2013 to 31-Dec-2018 under VCS program. The total GHG emission reductions generated in this monitoring period are 627,605 tCO₂e.

1.2 Sectoral Scope and Project Type

Sectoral scope: 1 energy industries (Renewable sources).

Project type: Grid connected wind power project.

The project is not a grouped project.

1.3 Project Proponent

Organization name	Shuangyashan Longyuan Wind Power Co., Ltd.
Contact person	Mr. Zhang Yulei
Title	Marketing Manager

Address	No.219, Nanzhi Road, Nangang District, Haerbin City, Heilongjiang Province,P.R. China
Telephone	+8645151919150
Email	yulei.zhang.c@chnenergy.com.cn

1.4 Other Entities Involved in the Project

Organization name	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.
Role in the Project	Consultancy
Contact person	Ma Qian
Title	General Manager
Address	1701-085, Floor 17, No.9 North 4 th Ring West Road, Haidian District, China
Telephone	+86 10 63887035
Email	e0068461@ceic.com

1.5 Project Start Date

25-Dec-2009 (started commissioning).

1.6 Project Crediting Period

The first 10-year crediting period of the project under VCS is from 25-Dec-2009 to 24-Dec-2019 (both days included).

1.7 Project Location

The project is located in Fengli Town, Jixian County, Shuangyashan City, Heilongjiang Province, and P.R.China. It is 23 kilometres away from Jixian County, and 30 kilometres away from Shuangyashan City. The center location's geographical coordinates are: East longitude 130°51'28'' (E130.85778°) and north latitude 46°44'35'' (N46.74306°), and altitude 240m~430m.

The detailed location of the project is shown in the Figure 1-1 below.



Figure 1-1 Location of the project

1.8 Title and Reference of Methodology

Applied methodology: ACM0002 (Version 12.1.0): Consolidated methodology for grid-connected electricity generation from renewable sources.

The methodology also refers to the following tools:

Tool to calculate the emission factor for an electricity system (Version 02.2.1)

Tool for the demonstration and assessment of additionality (Version 05.2.1)

The applied methodology and tools can be found from the following link:

<https://cdm.unfccc.int/methodologies/PAmethodologies/approved>

1.9 Participation under other GHG Programs

The project is a registered CDM project with reference No. 5491, the detailed information can be found at <https://cdm.unfccc.int/Projects/DB/BVQI1323445785.17/view>.

Start Date	End Date	GHG Emission Reduction (tCO ₂ e)	Credit	Program
25-Dec-2009	18-Dec-2011	218,614	VCU	VCS
19-Dec-2011	31-Dec-2012	102,399	CER	CDM

This is the second monitoring period under VCS program from 01-Jan-2013 to 31-Dec-2018 for the project. All emission reductions during this monitoring period have not and will not seek issuance under other kind of GHG programs. Emission reductions during this monitoring period will only seek issuance under VCS program.

1.10 Other Forms of Credit

Emission Trading Programs and Other Binding Limits

China has a national emissions trading scheme only cover the high-emission industries, such as thermal power generation, petrochemical, chemical, building materials, iron and steel, non-ferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons of CO₂e/year, not including renewable project¹.

The project proponent Shuangyashan Longyuan Wind Power Co., Ltd., as an enterprise for renewable energy investment, is not included in the compliance entity list by China national Emission Trading Scheme (ETS). Also, the project has not been registered as a CCER (Chinese Certified Emission Reductions) project in China, thus it is not eligible for emission reductions transaction

¹ http://www.mee.gov.cn/xxgk2018/xxgk/xxgk05/202103/t20210330_826728.html

under the China's ETS. Therefore, the project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading. The net GHG emission reductions generated during this monitoring period have not been used for compliance under such programs or mechanisms.

Furthermore, as a voluntary emission reduction project, the GHG emission reductions and removals generated by the project will not be enforced to be used for compliance in China. In addition, a statement on no double counting will be submitted to Verra to confirm the credits during this monitoring period has not been counted and will not be counted under emission trading programs and other binding limits.

Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates, during this monitoring period.

1.11 Sustainable Development Contributions

The project activity is a renewable wind power project located in Fengli Town, Jixian County, Shuangyashan City, Heilongjian Province, Northeast China during this monitoring period.



The project utilizes wind resources to generate and supply 610,513.279 MWh renewable electricity to SCPG, which contributes to SDG 7.



The project provides 13 long-term job opportunities for local, which has a positive effect on the local economy and contributes to SDG 8.



The project reduces 627,605 tCO₂e GHG emission and contributes to SDG 13.

These contribute to achieve China's stated sustainable development priorities, included strengthening resistance capacity to climate risks, increasing labor force participation rate, increasing the share of clean energy consumption.

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime ²
1)	7.2	Renewable energy supplied to the power grid	Implemented activities to increase	610,513.279 MWh of electricity from renewable sources has been supplied to the power grid during the reporting period	An accumulated 610,513.279 MWh of electricity from renewable sources has been supplied to the power grid by end of the reporting period
2)	8.5	Numbers of job opportunities provided by implementation of the project	Implemented activities to increase	13 people have been employed by the project as long-term employee	13 people have been employed by the project as long-term employee by end of the reporting period
3)	13.0	Tonnes of greenhouse gas emissions avoided	Implemented activities to increase	The project has achieved a total GHG emission reduction of 627,605 tCO ₂ e during the reporting period.	An accumulated 627,605 tCO ₂ e of GHG emission reduction have been achieved by the project by end of the reporting period

² The contributions over the project lifetime column are intended for displaying the cumulative SD contributions that have been accounted for and approved by Verra in an SD contributions report or VCS monitoring report. Since this is the first time project 1049 Heilongjiang Bianfushan Wind Power Project has claimed SD contributions, the contributions over the project lifetime should be the same as the current contributions for the period 01/01/2013-31/12/2018.

2 SAFEGUARDS

2.1 No Net Harm

Environmental Impact

The Environmental Impact Assessment (EIA) of the project has been conducted and approved by local government officials. The EIA report has assessed every possible aspect of environmental impacts of the project and recommended corresponding measures, where applicable. The environmental impacts and corresponding mitigation measures during operation have been discussed in the registered PD. No negative environmental impacts have been identified.

Socio-economic impact

Furthermore, the project is renewable project and achieves the SDG 7 and SDG 13. Also, the project provides 13 long-term job opportunities for local, which has a positive effect on the local economy and contributes to SDG 8. Please refer to section 1.11 of this report for details.

In conclusion, no net harm on local environment and social community has been detected for the project.

2.2 Local Stakeholder Consultation

LSC prior to the project implementation

Local stakeholder consultation has been conducted on 10-May-2009. All comments received have been well addressed prior to the project implementation. Please refer to section E of the registered PD (version 03 dated 01-Nov-2011) of the project for detailed information.

In conclusion, there was no significant negative comments received in the project preparation stage.

LSC during the operation period

Communications with Local stakeholders have been carried out at periodic intervals. Key implementation schedules and changes of the project have been communicated to the local authority, the neighbourhood committee, and local residents. The comments and suggestions from residents were collected by the local authority, if any.

The project owner carried out questionnaire survey for the local stakeholders to collect the relevant comments and suggestions every two years, e.g., in June 2014, July 2016 and May 2018 during this monitoring period. The local authority also conducts spot checks on the implementation of the project at periodic intervals as per relevant regulations. There have been no negative comments received for the project so far.

In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

2.3 AFOLU-Specific Safeguards

The project is a non-AFOLU project, this section is not required.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The project scenario is the implementation of the project, the installation and operation of 33 sets of wind turbines with a total capacity of 49.5MW which supply an average annual generation of 121,625MWh to NECPG and replace the same amount of electricity generated by fossil fuel fired power plants connected to NECPG. The type of wind turbine is GE77-1500. There is one 35kV line which gather the electricity generated from 33 sets of wind turbines to the cubicle switchboard located in the booster stations and the electricity would be sent to the substation. All wind turbines are produced in China. Technical Characteristics of Wind Turbines is shown as below table 3-1.

During this monitoring period, the project was implemented in accordance with the registered PD. The Project has operated without any accidental or emergency events that might impact the GHG emission reductions or removals and monitoring.

Table 3-1 Technical Characteristics of Wind Turbines³

³ Sourced from Wind Turbines Purchase Agreement

Parameters	Data
Rated wind speed	12m/s
Nominal output:	1500kW
Cut-in wind speed:	3m/s
Cut-out wind speed	22m/s
Survival wind speed	52.5m/s
Lifetime	20 Years
Manufacturer	General Electric (Shenyang) Co., Ltd.

3.2 Deviations

3.2.1 Methodology Deviations

There is no methodology deviation in this monitoring period.

3.2.2 Project Description Deviations

There is a deviation for the crediting period. The project is registered under VCS version 3.3 and completed validation before 19-Mar-2020, thus it remains eligible to apply the crediting period requirements under VCS Version 3.3 which shall be a maximum of ten years and may be renewed at most twice, so the first renewable crediting period of the project shall be updated from 25-Dec-2009 to 18-Dec-2011 to 25-Dec-2009~24-Dec-2019. This monitoring period belongs to the first crediting period from 25-Dec-2009~24-Dec-2019. The project did not complete the validation of the crediting period update within two years of the end of the first crediting period, therefore the VCS total crediting period ends on 24-Dec-2019, the end of the first crediting period.

The deviation does not impact the applicability of the methodology, additionality and the appropriateness of the baseline scenario.

3.3 Grouped Projects

Not applicable.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

The baseline grid emission factor $EF_{grid,CM,y}$ is calculated and fixed ex ante based on the relevant basis parameters in registered PD which are not described in detail here. With consideration of the fact of the Project, data and parameters that are available at validation are summarized in below table.

Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor for grid connected power generation in year y
Source of data	Registered PD
Value applied	1.0280
Justification of choice of data or description of measurement methods and procedures applied	Ex-ante determined in the registered PD and fixed during the 1 st crediting period.
Purpose of Data	Calculation of baseline emissions
Comments	Determined ex-ante and fixed for the 1 st crediting period

4.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Electricity meters reading on project activity site
Description of measurement methods and procedures to be applied	The monitoring meters (M1,M2,M3,M4 and M5) with accuracy of 0.5s. Four meters (M1, M2, M3, M4) were installed in 4 loops respectively, which were used to monitor electricity exported to the grid ($EG_{export,y}$) by the project. One meter (M5) were installed at the inlet of the main transformer, which were used to measure the electricity imported from the grid($EG_{import, y}$) by the project.

	Net electricity supplied by the project to the grid is the difference between the electricity exported to and the electricity imported from the grid.					
Frequency of monitoring/recording	The reading of electricity meters were continuously measured, daily read and monthly aggregated.					
Value monitored	Period		Value			
	01-Jan-2013~31-Dec-2013		113,788,860			
	01-Jan-2014~31-Dec-2014		94,888.572			
	01-Jan-2015~31-Dec-2015		80,566.227			
	01-Jan-2016~31-Dec-2016		94,670.530			
	01-Jan-2017~31-Dec-2017		103,750.398			
	01-Jan-2018~31-Dec-2018		122,848.692			
	Total		610,513.279			
Monitoring equipment	Monitoring equipment	M1	M2	M3	M4	M5
	Type	DSSD331	DSSD331	DSSD331	DSSD331	DSSD331
	Serial No.	09360003010500	09360003030500	09360003130500	09360003120500	09360003080500
	Accuracy	0.5s	0.5s	0.5s	0.5s	0.5s
	Calibration frequency	Annually	Annually	Annually	Annually	Annually
	Calibration date	13-Nov-2012 06-Nov-2013 28-Oct-2014 03-Jul-2015 02-Jul-2016 21-Jun-2017 16-Jun-2018	13-Nov-2012 06-Nov-2013 28-Oct-2014 03-Jul-2015 02-Jul-2016 21-Jun-2017 16-Jun-2018	13-Nov-2012 06-Nov-2013 28-Oct-2014 03-Jul-2015 02-Jul-2016 21-Jun-2017 16-Jun-2018	13-Nov-2012 06-Nov-2013 28-Oct-2014 03-Jul-2015 02-Jul-2016 21-Jun-2017 16-Jun-2018	13-Nov-2012 06-Nov-2013 28-Oct-2014 03-Jul-2015 02-Jul-2016 21-Jun-2017 16-Jun-2018
	Validity	12-Nov-2013	12-Nov-	12-Nov-	12-Nov-	12-Nov-

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	20-Jun-2018	2015	2015	2015	2015																																																																										
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		15-Jun-	15-Jun-	15-Jun-	15-Jun-																																																																										
		2019	2019	2019	2019																																																																										
QA/QC procedures to be applied	Electricity meters have been properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448). Accuracy degree of meters not less than 0.5s. Meter readings crosschecked with electricity purchase and sale receipts. Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup.																																																																														
Purpose of the data	Calculation of baseline emissions																																																																														
Calculation method	Net electricity supplied by the project to the grid is the difference between the electricity exported and the electricity imported.																																																																														
Comments	-																																																																														

Data / Parameter	EG _{export, y}
Data unit	MWh
Description	Electricity exported to the grid by the project
Source of data	Electricity meters M1, M2, M3 and M4 reading on project activity site
Description of measurement methods and procedures to be applied	The monitoring meters (M1,M2,M3 and M4) with accuracy of 0.5s were installed in 4 loops respectively.The electronic multifunctional and bidirectional meters can measure the electricity exported to the grid.
Frequency of monitoring/recording	The reading of electricity meters were continuously measured, daily read and monthly aggregated.

Value monitored	Period			Value	
	01-Jan-2013~31-Dec-2013			114,200.178	
	01-Jan-2014~31-Dec-2014			95,352.782	
	01-Jan-2015~31-Dec-2015			81,124.593	
	01-Jan-2016~31-Dec-2016			95,054.690	
	01-Jan-2017~31-Dec-2017			104,193.349	
	01-Jan-2018~31-Dec-2018			123,269.695	
	Total			613,195.287	
Monitoring equipment	Monitoring equipment	M1	M2	M3	M4
	Type	DSSD331	DSSD331	DSSD331	DSSD331
	Serial No.	09360003010500	09360003030500	09360003130500	09360003120500
	Accuracy	0.5s	0.5s	0.5s	0.5s
	Calibration frequency	Annually	Annually	Annually	Annually
	Calibration date	13-Nov-2012 06-Nov-2013 28-Oct-2014 03-Jul-2015 02-Jul-2016 21-Jun-2017 16-Jun-2018	13-Nov-2012 06-Nov-2013 28-Oct-2014 03-Jul-2015 02-Jul-2016 21-Jun-2017 16-Jun-2018	13-Nov-2012 06-Nov-2013 28-Oct-2014 03-Jul-2015 02-Jul-2016 21-Jun-2017 16-Jun-2018	13-Nov-2012 06-Nov-2013 28-Oct-2014 03-Jul-2015 02-Jul-2016 21-Jun-2017 16-Jun-2018
	Validity date	12-Nov-2013 05-Nov-2014 27-Oct-2015 02-Jul-2016 01-Jul-2017 20-June-2018 15-Jun-2019	12-Nov-2013 05-Nov-2014 27-Oct-2015 02-Jul-2016 01-Jul-2017 20-June-2018 15-Jun-2019	12-Nov-2013 05-Nov-2014 27-Oct-2015 02-Jul-2016 01-Jul-2017 20-June-2018 15-Jun-2019	12-Nov-2013 05-Nov-2014 27-Oct-2015 02-Jul-2016 01-Jul-2017 20-June-2018 15-Jun-2019
	QA/QC procedures to be applied				
	Electricity meters have been properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448). Accuracy degree of meters not less than 0.5s. Meter readings crosschecked with electricity purchase and sale receipts.				
	Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup.				
Purpose of the data		Calculation of baseline emissions			

Calculation method	-
Comments	-

Data / Parameter	EGimport, y	
Data unit	MWh	
Description	Electricity utilized by the project	
Source of data	Electricity meter M5 reading on project activity site.	
Description of measurement methods and procedures to be applied	The monitoring meter (M5) with accuracy of 0.5s installed at the inlet of the main transformer, which was used to measure the electricity imported from the grid(EGimport, y) by the project.	
Frequency of monitoring/recording	The reading of electricity meters were continuously measured, daily read and monthly aggregated.	
Value monitored	Period	Value
	01-Jan-2013~31-Dec-2013	411.318
	01-Jan-2014~31-Dec-2014	464.210
	01-Jan-2015~31-Dec-2015	558.366
	01-Jan-2016~31-Dec-2016	384.160
	01-Jan-2017~31-Dec-2017	442.951
	01-Jan-2018~31-Dec-2018	421.003
	Total	2,682.008
Monitoring equipment	Monitoring equipment	M5
	Type	DSSD331
	Serial No.	09360003080500
	Accuracy	0.5s
	Calibration frequency	Annually
	Calibration date	13-Nov-2012 06-Nov-2013 28-Oct-2014 03-Jul-2015 02-Jul-2016

		21-Jun-2017 16-Jun-2018
	Validity date	12-Nov-2013 05-Nov-2014 27-Oct-2015 02-Jul-2016 01-Jul-2017 20-June-2018 15-Jun-2019
QA/QC procedures to be applied	Electricity meters have been properly calibrated annually according to the requirement from Technical administrative code of electric energy metering (DL/T448). Accuracy degree of meters not less than 0.5s. Meter readings crosschecked with electricity purchase and sale receipts. Data will be archived for 2 years following the end of the crediting period by means of electronic and paper backup.	
Purpose of the data	Calculation of baseline emissions	
Calculation method	-	
Comments	-	

4.3 Monitoring Plan

Organizational structure

The organizational structure and responsibilities of the personnel is shown as follows:

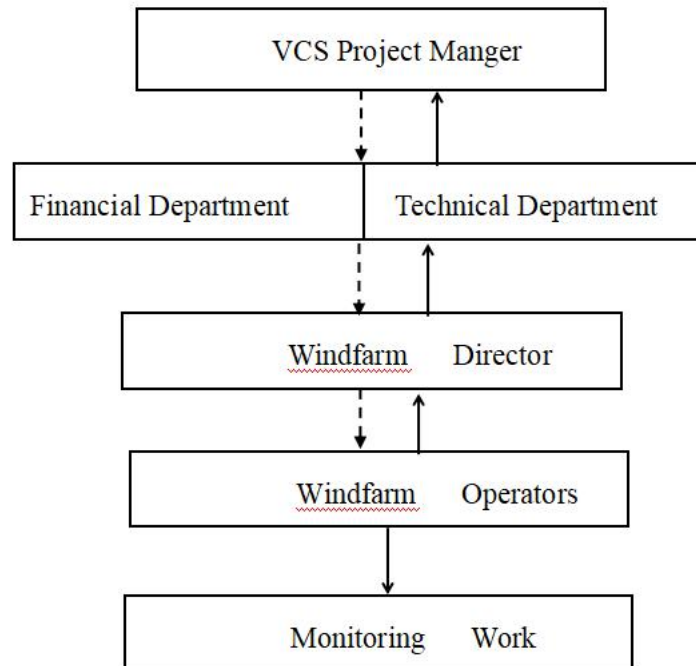


Figure 4-1 Organizational structure

Windfarm Derector collects the information and data required by the Monitoring Plan. The collected information should be documented and sent to the VCS manager and responsible staffs of the Shuangyashan Longyuan Wind Power Co., Ltd. monthly.

The VCS manager is in charge of the implementation of the Monitoring Plan and the confirmations on monitoring, calculation data and report to the General Manager of the company. Managers of the project must maintain credible, transparent, and adequate data estimation, measurement, collection, and tracking systems to maintain the information required for an audit of an emission reduction project.

Monitoring equipment

For the project, all the wind turbines are divided into 4 loops to parallel-connect to main transformer. Four meters (M1, M2, M3, M4) are installed in 4 loops respectively, which are used to monitor electricity exported to the grid ($EG_{export,y}$) by the project. The meter M5 installed at the inlet of the main transformer is used to monitor electricity utilized by the project ($EG_{import,y}$).

A technology diagram of the project is presented as Figure 4-2.

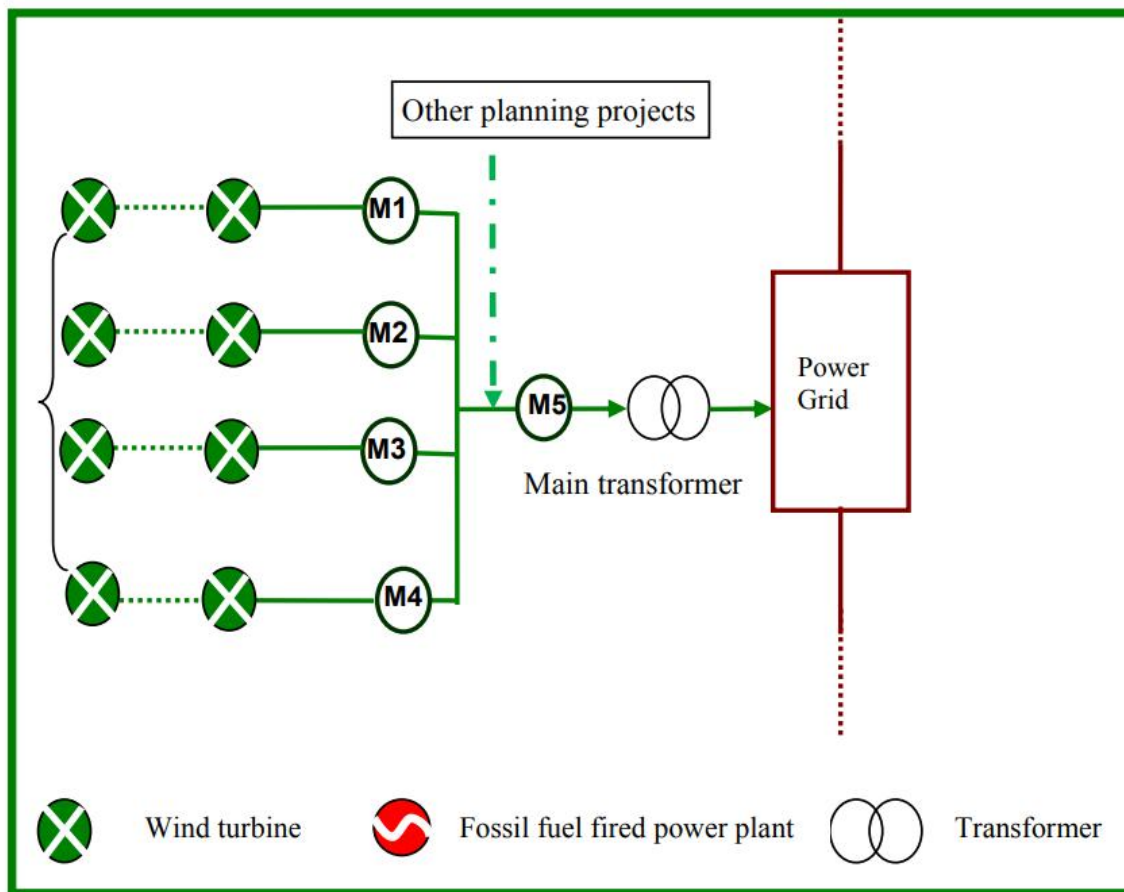


Figure 4-2 A technology diagram of the project

Data collection

The results from the meters are recorded by the designated persons of the project at 24:00 on the last day of every month and reported to the grid company. After the Grid Company reads the meters, and confirms the reading with the project owner, the electricity sales receipts based on the checked electricity amount are issued.

QA/QC and emergency procedure

Shuangyashan Longyuan Wind Power Co., Ltd specially issued the regulations of monitoring and management for VCS project to monitor and verify the emission reductions from the project. Moreover, the management procedures and regulations like operation, meter recordings, maintenance and emergency management are established in the VCS Manual, which guides the technicians to operate and guarantee the quality assurance and quality control procedures for recording, maintaining and archiving data in detail. This is an on-going process that will be ensured through the VCS in terms of the need for

verification of the emissions on an annual basis according to methodology ACM0002. Several processes required to meet the requirements for emissions reduction monitoring include:

- The project owner reads the meters system monthly and supplies the recordings to the local Electric Power Company. The local Electric Power Company checks the recordings supplied by the project owner.
- If either party finds any reading of the Meters more than the allowable error or the monitoring equipment functions improperly, it should inform the other party immediately. The project owner and the local Power Grid Company should retain a qualified measure institute together to check the meters or equipment, solve the problems and get everything into normal condition; In the mean time, both sides shall jointly prepare an acceptable estimation method according to Regulations and Rules of Power Supply issued by State Electricity Regulatory Commission of China to calculate the electricity exported to the Grid, and provide evidence to VVB for the verification to show the estimation is reasonable and conservative.

For the handling of disputes between the project owner and the Grid, measures will be adopted according to relevant articles of Interim Measures for Settlement of Electricity Charges between the Power Generating Enterprises and the Grid Enterprises 48 issued by State Electricity Regulatory Commission of China.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

As per the registered PD, the baseline emission of the project is calculated as below:

$$BE_y = EG_{\text{facility},y} \times EF_{\text{grid,CM},y} \quad (1)$$

Where,

BE_y Baseline emissions in year y (t CO₂e)

$EG_{facility,y}$ Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)

$EF_{grid,CM,y}$ Combined margin CO₂ emission factor for grid connected power generation in year y (tCO₂e/MWh)

The monitored $EG_{facility,y}$ is reported in the table below.

Table 5-1 Monitored monthly value of $EG_{\text{facility},y}$

Period		Monthly Recording output Data (=M1+M2+M3+M4)	Sales Receipt	$EG_{\text{export},y}$	Monthly Recording input Data (M5)	Sales Receipt	$EG_{\text{import},y}$	$EG_{\text{facility},y}$
		A	B	$C=\min(A, B)$	D	E	$F=\max(D, E)$	$G=C-F$
		MWh	MWh	MWh	MWh	MWh	MWh	MWh
01-Jan-2013	31-Jan-2013	11,601.934	11,601.934	11,601.934	49.638	49.638	49.638	11,552.296
01-Feb-2013	28-Feb-2013	11,029.041	11,029.041	11,029.041	0	0	0	11,029.041
01-Mar-2013	31-Mar-2013	15,389.191	15,389.191	15,389.191	0	0	0	15,389.191
01-Apr-2013	30-Apr-2013	9,533.908	9,533.908	9,533.908	33.000	33.000	33.000	9,500.908
01-May-2013	31-May-2013	6,431.434	6,431.434	6,431.434	22.176	22.176	22.176	6,409.258
01-Jun-2013	30-Jun-2013	6,628.514	6,628.514	6,628.514	37.488	37.488	37.488	6,591.026
01-Jul-2013	31-Jul-2013	5,882.152	5,882.152	5,882.152	47.784	47.784	47.784	5,834.368
01-Aug-2013	31-Aug-2013	4,702.778	4,702.778	4,702.778	63.624	63.624	63.624	4,639.154
01-Sep-2013	30-Sep-2013	11,467.502	11,467.502	11,467.502	74.712	74.712	74.712	11,392.790
01-Oct-2013	31-Oct-2013	11,075.745	11,075.745	11,075.745	0	0	0	11,075.745
01-Nov-2013	30-Nov-2013	11,744.285	11,744.285	11,744.285	82.896	82.896	82.896	11,661.389
01-Dec-2013	31-Dec-2013	8,713.694	8,713.694	8,713.694	0	0	0	8,713.694
Subtotal-2013		114,200.178	114,200.178	114,200.178	411.318	411.318	411.318	113,788.860
01-Jan-2014	31-Jan-2014	9,262.576	9,262.576	9,262.576	44.616	44.616	44.616	9,217.960
01-Feb-2014	28-Feb-2014	5,499.397	5,499.397	5,499.397	65.736	65.736	65.736	5,433.661
01-Mar-2014	31-Mar-2014	7,486.252	7,486.252	7,486.252	40.939	40.939	40.939	7,445.313
01-Apr-2014	30-Apr-2014	7,488.936	7,488.936	7,488.936	34.344	34.344	34.344	7,454.592

01-May-2014	31-May-2014	9,693.706	9,693.706	9,693.706	19.811	19.811	19.811	9,673.895
01-Jun-2014	30-Jun-2014	5,112.785	5,112.785	5,112.785	35.683	35.683	35.683	5,077.102
01-Jul-2014	31-Jul-2014	7,881.740	7,881.740	7,881.740	50.160	50.160	50.160	7,831.580
01-Aug-2014	31-Aug-2014	3,437.646	3,437.646	3,437.646	31.680	31.680	31.680	3,405.966
01-Sep-2014	30-Sep-2014	6,585.214	6,585.214	6,585.214	38.280	38.280	38.280	6,546.934
01-Oct-2014	31-Oct-2014	13,055.977	13,055.977	13,055.977	22.440	22.440	22.440	13,033.537
01-Nov-2014	30-Nov-2014	11,231.727	11,231.727	11,231.727	80.521	80.521	80.521	11,151.206
01-Dec-2014	31-Dec-2014	8,616.826	8,616.826	8,616.826	0	0	0	8,616.826
Subtotal-2014		95,352.782	95,352.782	95,352.782	464.210	464.210	464.210	94,888.572
01-Jan-2015	31-Jan-2015	10,326.916	10,326.916	10,326.916	35.640	35.640	35.640	10,291.276
01-Feb-2015	28-Feb-2015	3,730.495	3,730.495	3,730.495	35.640	35.640	35.640	3,694.855
01-Mar-2015	31-Mar-2015	4,452.577	4,452.577	4,452.577	130.681	130.681	130.681	4,321.896
01-Apr-2015	30-Apr-2015	11,570.093	11,570.093	11,570.093	40.920	40.920	40.920	11,529.173
01-May-2015	31-May-2015	8,822.057	8,822.057	8,822.057	30.360	30.360	30.360	8,791.697
01-Jun-2015	30-Jun-2015	6,863.942	6,863.942	6,863.942	33.000	33.000	33.000	6,830.942
01-Jul-2015	31-Jul-2015	5,508.595	5,508.595	5,508.595	44.880	44.880	44.880	5,463.715
01-Aug-2015	31-Aug-2015	4,519.103	4,519.103	4,519.103	66.000	66.000	66.000	4,453.103
01-Sep-2015	30-Sep-2015	3,817.300	3,817.300	3,817.300	42.240	42.240	42.240	3,775.060
01-Oct-2015	31-Oct-2015	6,882.375	6,882.375	6,882.375	19.800	19.800	19.800	6,862.575
01-Nov-2015	30-Nov-2015	4,714.757	4,714.757	4,714.757	33.003	33.003	33.003	4,681.754
01-Dec-2015	31-Dec-2015	9,916.383	9,916.383	9,916.383	46.202	46.202	46.202	9,870.181
Subtotal-2015		81,124.593	81,124.593	81,124.593	558.366	558.366	558.366	80,566.227
01-Jan-2016	31-Jan-2016	11,378.992	11,378.992	11,378.992	47.522	47.522	47.522	11,331.470
01-Feb-2016	29-Feb-2016	3,911.989	3,911.989	3,911.989	31.684	31.684	31.684	3,880.305

01-Mar-2016	31-Mar-2016	4,661.811	4,661.811	4,661.811	47.522	47.522	47.522	4,614.289
01-Apr-2016	30-Apr-2016	9,466.403	9,466.403	9,466.403	18.481	18.481	18.481	9,447.922
01-May-2016	31-May-2016	13,364.900	13,364.900	13,364.900	15.841	15.841	15.841	13,349.059
01-Jun-2016	30-Jun-2016	3,478.568	3,478.568	3,478.568	56.767	56.767	56.767	3,421.801
01-Jul-2016	31-Jul-2016	4,304.965	4,304.965	4,304.965	0	0	0	4,304.965
01-Aug-2016	31-Aug-2016	6,785.902	6,785.902	6,785.902	68.653	68.653	68.653	6,717.249
01-Sep-2016	30-Sep-2016	8,495.996	8,495.996	8,495.996	39.605	39.605	39.605	8,456.391
01-Oct-2016	31-Oct-2016	14,457.612	14,457.612	14,457.612	11.880	11.880	11.880	14,445.732
01-Nov-2016	30-Nov-2016	8,769.675	8,769.675	8,769.675	0	0	0	8,769.675
01-Dec-2016	31-Dec-2016	5,977.877	5,977.877	5,977.877	46.205	46.205	46.205	5,931.672
Subtotal-2016		95,054.690	95,054.690	95,054.690	384.160	384.160	384.160	94,670.530
01-Jan-2017	31-Jan-2017	6,577.288	6,577.288	6,577.288	21.121	21.121	21.121	6,556.167
01-Feb-2017	28-Feb-2017	6,121.437	6,121.437	6,121.437	26.401	26.401	26.401	6,095.036
01-Mar-2017	31-Mar-2017	4,132.075	4,132.075	4,132.075	46.207	46.207	46.207	4,085.868
01-Apr-2017	30-Apr-2017	12,024.665	12,024.665	12,024.665	30.363	30.363	30.363	11,994.302
01-May-2017	31-May-2017	14,444.109	14,444.109	14,444.109	18.499	18.499	18.499	14,425.610
01-Jun-2017	30-Jun-2017	6,765.948	6,765.948	6,765.948	42.246	42.246	42.246	6,723.702
01-Jul-2017	31-Jul-2017	5,899.638	5,899.638	5,899.638	56.777	56.777	56.777	5,842.861
01-Aug-2017	31-Aug-2017	8,360.670	8,360.670	8,360.670	59.412	59.412	59.412	8,301.258
01-Sep-2017	30-Sep-2017	9,009.993	9,009.993	9,009.993	31.685	31.685	31.685	8,978.308
01-Oct-2017	31-Oct-2017	12,378.854	12,378.854	12,378.854	26.402	26.402	26.402	12,352.452
01-Nov-2017	30-Nov-2017	11,454.595	11,454.595	11,454.595	31.585	31.585	31.585	11,423.010
01-Dec-2017	31-Dec-2017	7,024.077	7,024.077	7,024.077	52.253	52.253	52.253	6,971.824
Subtotal-2017		104,193.349	104,193.349	104,193.349	442.951	442.951	442.951	103,750.398

01-Jan-2018	31-Jan-2018	13,712.357	13,712.357	13,712.357	22.993	22.993	22.993	13,689.364
01-Feb-2018	28-Feb-2018	7,412.939	7,412.939	7,412.939	31.681	31.681	31.681	7,381.258
01-Mar-2018	31-Mar-2018	13,080.583	13,080.583	13,080.583	14.521	14.521	14.521	13,066.062
01-Apr-2018	30-Apr-2018	10,629.741	10,629.741	10,629.741	59.406	59.406	59.406	10,570.335
01-May-2018	31-May-2018	7,720.481	7,720.481	7,720.481	27.724	27.724	27.724	7,692.757
01-Jun-2018	30-Jun-2018	7,241.044	7,241.044	7,241.044	33.007	33.007	33.007	7,208.037
01-Jul-2018	31-Jul-2018	6,507.290	6,507.290	6,507.290	35.649	35.649	35.649	6,471.641
01-Aug-2018	31-Aug-2018	4,090.419	4,090.419	4,090.419	76.647	76.647	76.647	4,013.772
01-Sep-2018	30-Sep-2018	9,750.385	9,750.385	9,750.385	44.984	44.984	44.984	9,705.401
01-Oct-2018	31-Oct-2018	13,240.657	13,240.657	13,240.657	42.358	42.358	42.358	13,198.299
01-Nov-2018	30-Nov-2018	12,681.646	12,681.646	12,681.646	19.945	19.945	19.945	12,661.701
01-Dec-2018	31-Dec-2018	17,202.153	17,202.153	17,202.153	12.088	12.088	12.088	17,190.065
Subtotal-2018		123,269.695	123,269.695	123,269.695	421.003	421.003	421.003	122,848.692
Total		613,195.287	613,195.287	613,195.287	2,682.008	2,682.008	2,682.008	610,513.279

The annual baseline emissions of the project are calculated as follows:

Table 5-2 Calculation of Baseline emissions

Year	EG _{facility,y} (MWh)	EF _{grid,CM,y} (tCO ₂ /MWh)	BE _y (tCO ₂)
	A	B	C=A*B
01-Jan-2013~31-Dec-2013	113,788,860	1.0280	116,974
01-Jan-2014~31-Dec-2014	94,888.572	1.0280	97,545
01-Jan-2015~31-Dec-2015	80,566.227	1.0280	82,822
01-Jan-2016~31-Dec-2016	94,670.530	1.0280	97,321

01-Jan-2017~31-Dec-2017	103,750.398	1.0280	106,655
01-Jan-2018~31-Dec-2018	122,848.692	1.0280	126,288
Total	610,513,279	-	627,605

In summary, the baseline emissions BE_y during this monitoring period is 627,605tCO₂e.

5.2 Project Emissions

According to the ACM0002, the emission of wind power project activity is zero, i.e. $PE_y=0$.

5.3 Leakage

According to ACM0002, the leakage of wind power project is not needed to be considered.

5.4 Net GHG Emission Reductions and Removals

This monitoring period started from 01-Jan-2013 to 31-Dec-2018, with totally 2,191 days. Based on the annual estimated emission reductions from the registered PD, the amount of emission reductions for this monitoring period would be $125,030 \text{ tCO}_2\text{e}/365 \text{ d} \times 2,191 \text{ d} = 750,523 \text{ tCO}_2\text{e}$. The actual emission reductions in this monitoring period (2,191 days) are 627,605 tCO₂e, which are 16.38% lower than the estimation in the registered PD. It is because wind fluctuations and the limitation of the grid absorption capacity. So, it is reasonable that during this monitoring period, the actual emission reduction is less than the estimated value.

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
01-Jan-2013~31-Dec-2013	116,974	0	0	116,974
01-Jan-2014~31-Dec-2014	97,545	0	0	97,545
01-Jan-2015~31-Dec-2015	82,822	0	0	82,822
01-Jan-2016~31-Dec-2016	97,321	0	0	97,321
01-Jan-2017~31-Dec-2017	106,655	0	0	106,655

01-Jan-2018~31-Dec-2018	126,288	0	0	126,288
Total	627,605	0	0	627,605

APPENDIX1: <SUPPORTING EVIDENCE>

Eviednce for SDG7 and SDG13:

The electricity supplied during this monitoring period between 01-Jan-2013 to 31-Dec-2018 is 610,513.279 MWh. The total emission reductions claimed in this monitoring period are 627,605 tCO₂e. and the emission reduction generated from the project during this monitoring period is 627,605 tCO₂e as mentioned in section 5.4 above, which is calculated from meter(s) readings and Sales receipts provided by the project owner, which will be verified by VVB.

Evidence for SDG 8: The staff payroll of the project has been submitted as a separate document to Verra due to confidentiality requirement of the project proponent.